Introduction to Computer Science

Introductory study guide · Prepared as sample educational content for the StudentHub frontend project.

1. What is computer science?

Computer science is the study of computation: how information is represented, processed, stored, and communicated. It includes programming, algorithms, computer systems, networks, security, and the social effects of technology.

2. Data and representation

Computers represent information using bits (0s and 1s). Groups of bits can encode numbers, text, images, sound, and instructions. The same bit pattern can mean different things depending on the agreed encoding.

3. Hardware and software

Hardware is the physical equipment: processor, memory, storage, and input/output devices. Software is the set of programs and instructions that tell hardware what tasks to perform. An operating system coordinates resources and provides common services.

4. Algorithms

An algorithm is a finite, ordered set of steps for solving a problem. A useful algorithm has clearly defined inputs and outputs, produces the intended result, and eventually finishes. Efficiency considers resources such as time and memory.

5. Networks and the internet

A computer network connects devices so they can exchange data. The internet is a global network of networks. Protocols define how devices address, format, transmit, and interpret information.

6. Responsible computing

Technology choices affect privacy, accessibility, security, and the environment. Good computing practice includes protecting data, checking sources, designing for different users, and considering who may be excluded or harmed.

Quick review

Explain the difference between hardware and software. Give an example of an algorithm from daily life. Why do networks need protocols? Name two responsibilities of a software developer.